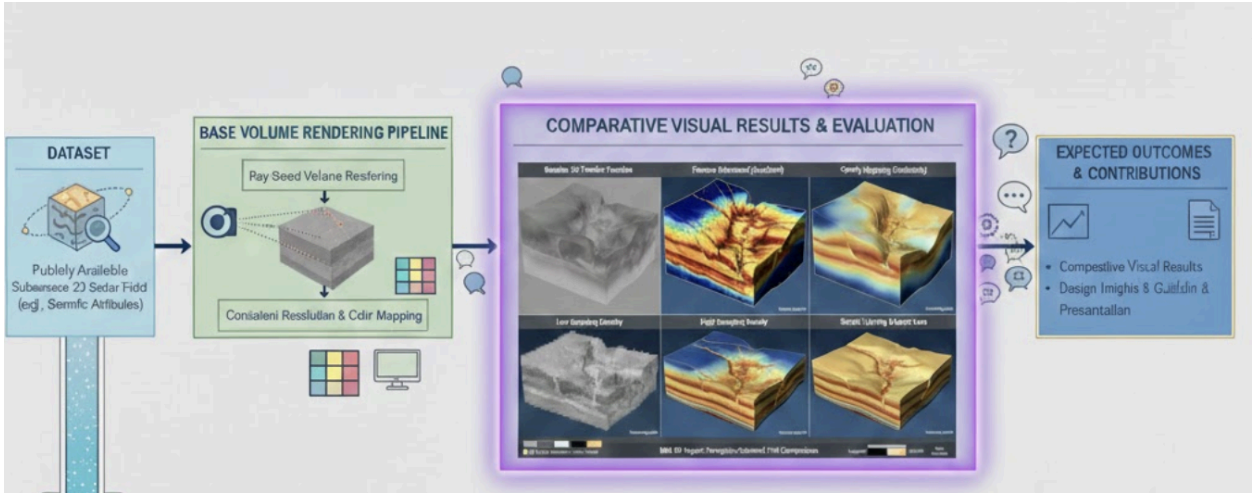
MM 804 Course Project Proposal

Perception-Oriented Exploration of Direct Volume Rendering for Subsurface Geophysical Data



This figure summarizes our workflow. We start with a public subsurface 3D scalar dataset, apply a standard direct volume rendering pipeline, and then systematically compare different visualization design choices such as transfer functions. The final outcome is a set of side-by-side visual results and design insights that explain which configurations improve interpretability.

Research Gap

Subsurface and geophysical data are inherently 3D, but in practice they are still mostly explored using slices or simple surfaces.

Although volume rendering exists, there is little discussion about *why* certain volume rendering settings actually make subsurface structures easier or harder to understand.

In other words, the visualization design choices are often used, but not systematically compared from a perception point of view.

We know volume rendering works, but we don’t clearly understand which visualization choices actually help people interpret subsurface structures.

What we plan to do

We will use direct volume rendering on a publicly available subsurface dataset and systematically compare visualization design choices.

Specifically, we will keep the data and viewpoint fixed, and vary things like transfer functions, sampling density, and basic shading to see how they affect structure visibility and depth perception.

The focus is on visualization design and interpretability, not on developing new rendering algorithms or doing domain analysis.

We’re doing a controlled, visualization-focused comparison to understand how volume rendering design choices affect interpretability.

Deliverables

By the end of the course, we will deliver:

1. A working volume rendering visualization of a subsurface dataset

2. A set of side-by-side visual comparisons showing different rendering configurations

3. A final presentation and report explaining the visualization design choices and what we learned

A working visualization, clear comparison results, and an explanation of the design trade-offs.

The final outcome is a set of volume-rendered visualizations that clearly demonstrate how visualization design choices affect the interpretability of subsurface structures.

1. Background and Motivation

Subsurface and geophysical datasets are inherently three-dimensional scalar fields, commonly representing physical properties such as seismic amplitude, density, or resistivity. These datasets play a critical role in understanding underground structures and spatial continuity.

Despite their volumetric nature, subsurface data are still predominantly explored using slice-based visualizations, such as inline and crossline sections, or simplified isosurfaces. While these approaches are intuitive and computationally efficient, they often fragment spatial context and obscure internal structures, making it difficult to perceive depth, continuity, and overlapping features.

Direct volume rendering (DVR) has long been established as a standard technique for visualizing 3D scalar fields, as it preserves continuous spatial information without requiring explicit surface extraction. However, in subsurface visualization practice, volume rendering is often treated as a presentation tool rather than an exploratory one, with limited discussion of how visualization design choices influence interpretability.

This project is motivated by the need to better understand how volume rendering can be used as an **exploratory visualization technique** for subsurface data, rather than merely a visually appealing rendering method.

2. Literature-Grounded Gap

Existing scientific visualization literature establishes direct volume rendering as a mature and widely adopted approach for rendering volumetric data. Prior research extensively explores rendering algorithms, acceleration techniques, and performance optimization.

However, two important gaps emerge when examining subsurface and geophysical visualization work:

1. **Emphasis on outcomes over perception**

   Many applications demonstrate volume-rendered results without systematically examining *why* certain visual configurations improve understanding. Visualization design choices—such as transfer function selection, sampling density, and basic shading—are often treated as ad hoc or experience-driven decisions.

2. **Lack of structured design comparison in non-medical domains**

   While medical imaging has a rich body of work discussing transfer function design and perceptual effects, subsurface visualization lacks structured, side-by-side comparisons that explain how different volume rendering configurations affect feature visibility, occlusion, and depth perception.

As a result, there is a gap in perception-oriented, design-focused explorations that clarify how direct volume rendering parameters support or hinder the interpretability of subsurface volumetric data.

3. Project Objective

The objective of this project is to **systematically explore how visualization design choices in direct volume rendering influence perceptual understanding of subsurface 3D scalar fields**.

Rather than proposing new rendering algorithms, this project focuses on visualization design and interpretability, aligning with the goals of scientific visualization and the scope of MM 804.

4. Dataset

The project will use a **publicly available subsurface or geophysical 3D scalar dataset**, such as seismic attributes, density fields, or resistivity volumes.

Dataset characteristics:

- Three-dimensional volumetric grid (e.g., x \times y \times z)
- Single scalar value per voxel
- Non-medical and non-disaster-related
- Not previously used in any coursework or research by the project group

The dataset selection emphasizes suitability for volume rendering and exploratory visualization rather than domain-specific scientific analysis.

5. Proposed Methodology

5.1 Baseline Volume Rendering Pipeline

A standard direct volume rendering pipeline will be implemented using an existing visualization framework (e.g., Blender or VTK-based tools). The pipeline includes:

- Ray-based volume rendering
- Fixed camera viewpoints to ensure fair visual comparison
- Consistent rendering resolution and color mapping

This baseline serves as a controlled foundation for comparative exploration.

5.2 Visualization Design Exploration

The core of the project is a **controlled comparison of visualization design parameters**, with one variable adjusted at a time:

1. **Transfer Function Design**
    - Baseline 1D intensity-based transfer function
    - Feature-enhanced transfer function (e.g., emphasizing boundaries or gradients)
    - Alternative opacity mappings highlighting internal continuity
2. **Sampling Density**
    - Low, medium, and high sampling rates
    - Examination of noise, smoothness, and structural continuity
3. **Basic Shading and Depth Cues**
    - No shading versus simple lighting models
    - Impact on depth perception and spatial understanding

5.3 Evaluation Strategy

Evaluation will be **qualitative and perception-oriented**, consistent with established practice in scientific visualization:

- Side-by-side visual comparisons under identical viewpoints
- Assessment criteria:
    - Visibility of internal structures
    - Depth perception and spatial continuity
    - Occlusion ambiguity
    - Overall interpretability for exploratory tasks

The goal is not numerical accuracy but visual clarity and explainability.

6. Expected Outcomes and Contributions

This project is expected to produce:

- A working direct volume rendering visualization of subsurface data
- A structured set of comparative visual results illustrating the effects of key rendering parameters
- Clear, explainable insights into how visualization design choices influence interpretability

- Practical visualization guidelines for exploratory volume rendering of subsurface scalar fields

These contributions are educational, visualization-focused, and well-aligned with the learning objectives of MM 804.

7. Deliverables

By the end of the course, the following deliverables will be produced:

1. A functional volume-rendered visualization of a subsurface dataset

2. A collection of comparative visualizations demonstrating different design choices

3. A final presentation explaining the methodology, findings, and design rationale

4. A concise written report documenting the visualization process and qualitative evaluation

All deliverables emphasize visualization design and graphical reasoning rather than domain-specific scientific conclusions.

8. Timeline and Feasibility

- **February**

  Dataset selection, preprocessing, and baseline volume rendering

- **March**

  Transfer function and parameter exploration; comparative visualization

- **April**

  Final visualization refinement and presentation preparation

The project scope is intentionally controlled to ensure feasibility within the course timeline while allowing meaningful exploration.

9. Alignment with Course Goals

This project aligns strongly with MM 804 by focusing on:

- Scientific visualization principles
- Volume rendering techniques
- Visualization design and perceptual considerations

- Clear, interpretable graphical outcomes

It avoids scope creep into algorithm development, prediction, or domain analysis, ensuring a focused and achievable course project.